

Question ID: 869a32f1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question

The high temperature, in degrees Fahrenheit (°F), in a certain city was recorded for each of 5 days. The data are shown below.

Day	1	2	3	4	5
High temperature (°F)	81	80	81	81	82

Over this 5-day period, which of the following is NOT equal to 81°F?

Answer

- A. Median of the high temperatures
- B. Mean of the high temperatures
- C. Mode of the high temperatures
- D. Range of the high temperatures

Correct Answer: D

Rationale

Choice D is correct. The range of a data set is the difference between the maximum and the minimum values in the set. The maximum value among the high temperatures in the table is 82°F and the minimum value is 80°F. Therefore, the range is $82^{\circ}\text{F} - 80^{\circ}\text{F} = 2^{\circ}\text{F}$.

Choice A is incorrect. The median of a data set is the middle value when the values in the set are ordered from least to greatest. Ordering the high temperatures this way gives the list 80, 81, 81, 81, 82. Therefore, the median high temperature is 81°F. Choice B is incorrect. The mean high temperature is $\frac{81+80+81+81+82}{5} = \frac{405}{5} = 81$. Choice C is incorrect. The mode is the value that occurs the greatest number of times. For the set of high temperatures shown, 81 is the value that occurs 3 times, and therefore, 81°F is the mode of the high temperatures.